

BMS\_SLAVE Fabrication Document

Layer Stack Legend

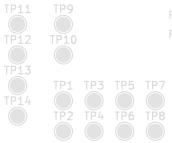
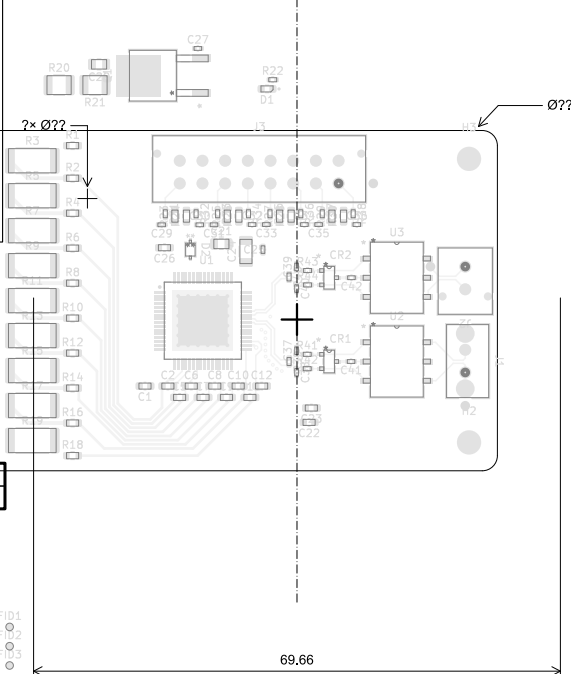
| Material     | Layer | Thickness        | Dielectric | Type          | Gerber |
|--------------|-------|------------------|------------|---------------|--------|
| F,Paste      |       |                  |            | Paste Mask    |        |
| F,Silkscreen |       |                  |            | Legend        | GBR    |
| F,Mask       |       | 0,02mm           |            | Solder Mask   | GBR    |
| Copper       | L1    | 0,035mm (1.00oz) |            | Signal        | GBR    |
| Prepreg      |       | 0,1855mm         | FR4_7628   | Dielectric    |        |
| Copper       | L2    | 0,035mm (1.00oz) |            | Signal        | GBR    |
| Core         |       | 1,03mm           | FR4        | Dielectric    |        |
| Copper       | L3    | 0,035mm (1.00oz) |            | Signal        | GBR    |
| Prepreg      |       | 0,1855mm         | FR4_7628   | Dielectric    |        |
| Copper       | L4    | 0,035mm (1.00oz) |            | Signal        | GBR    |
| B,Mask       |       | 0,02mm           |            | Solder Resist | GBR    |
| B,Silkscreen |       |                  |            | Legend        | GBR    |
| B,Paste      |       |                  |            | Paste Mask    |        |

Total thickness: 1,581mm  
Note: external layer thicknesses are specified after plating

Impedance Table

| Transmission Line              | Impedance [ohms] | Tolerance [ohms] | Layer | Trace Width [mm] | Gap [mm] | Ref. Layers |
|--------------------------------|------------------|------------------|-------|------------------|----------|-------------|
| Edge-Coupled Coated Microstrip | 100              | ±10 %            | L1    | 0,2032           | 0,28     | L2          |

Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge\_Cuts.GBR" SUFFIX.  
  
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.  
  
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BLUE DEVIL MOTOR SPORTS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE75.000 × 45.000 mm

BOARD THICKNESS1.581 mm

TRACE WIDTH0.200 mm

TRACE TO TRACE-0.000 mm

MIN. HOLE (PTH)0.200 mm

MIN. HOLE (NPTH)1.050 mm

ANNULAR RING0.150 mm

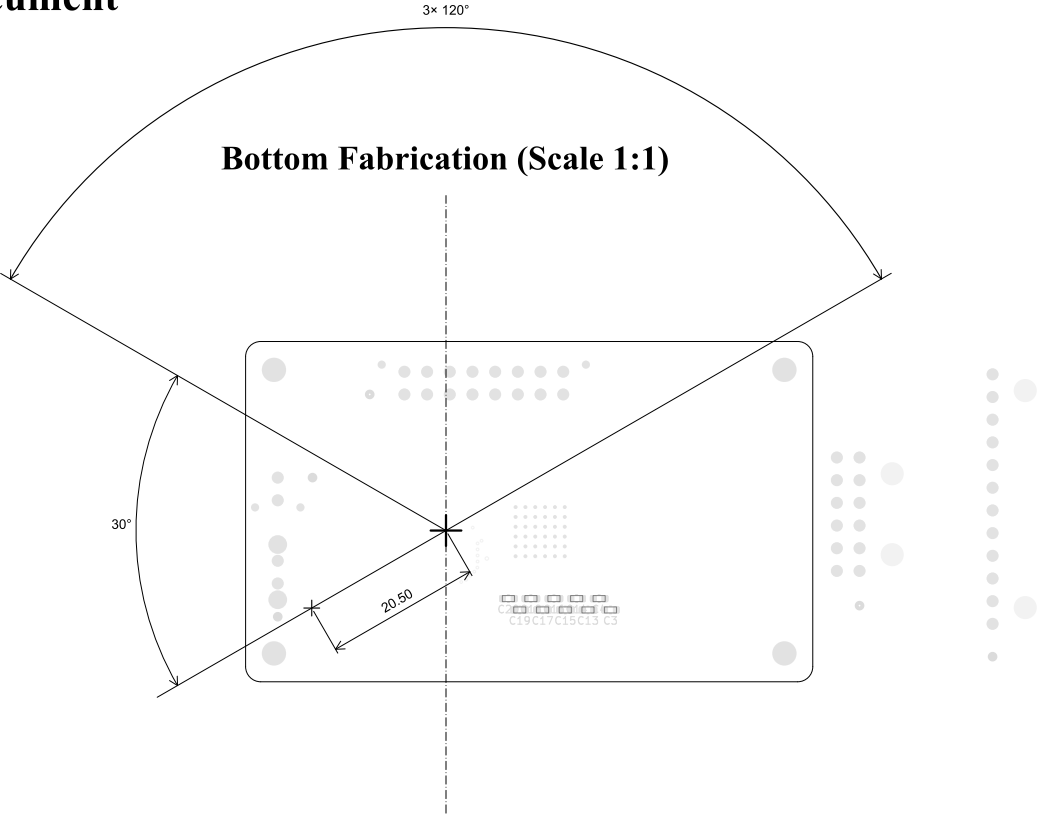
COPPER TO HOLE0.254 mm

COPPER TO EDGE0.250 mm

HOLE TO HOLE0.254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

|  |                                             |                                     |                               |                            |                             |
|--|---------------------------------------------|-------------------------------------|-------------------------------|----------------------------|-----------------------------|
|  | Comments:                                   | Company:<br>Blue Devil Motor Sports |                               | Variant:<br>PRELIMINARY    | Git Hash:<br>cd3370c        |
|  | Sheet Title:<br>Top Fabrication (Scale 1:1) | Board Name:<br>BMS_SLAVE            |                               | Project Name:<br>BMS Slave |                             |
|  | Sheet Path:                                 | File Name:<br>bms_slave.kicad_pcb   | Designer:<br>Aidan Brzezinski | Date:<br>2024-04-13        | Revision:<br>+ (Unreleased) |
|  |                                             |                                     | Reviewer:                     | Size:<br>A4                | Sheet:<br>1 of 10           |

BMS\_SLAVE Fabrication Document



All dimensions are in millimeters unless otherwise specified.

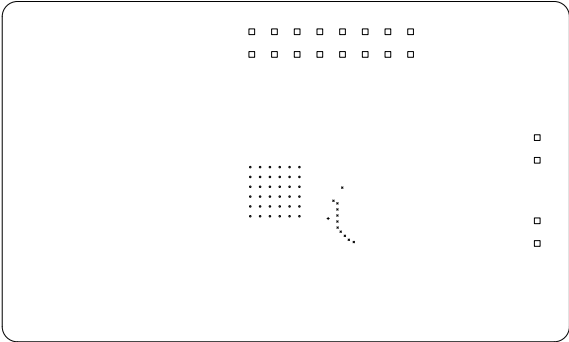
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|--------------------------------------------------------------------------------------|-----------|---------------------|-------------------------|------------------|-------------|---------------|-----------|----------------|--|
|  | Comments: |                     | Company:                |                  | Variant:    |               | Git Hash: |                |  |
|                                                                                      |           |                     | Blue Devil Motor Sports |                  | PRELIMINARY |               | cd3370c   |                |  |
|                                                                                      |           |                     | Board Name:             |                  |             | Project Name: |           |                |  |
|                                                                                      |           |                     | BMS_SLAVE               |                  |             | BMS Slave     |           |                |  |
| Sheet Title:                                                                         |           | File Name:          |                         | Designer:        |             | Date:         |           | Revision:      |  |
| Bottom Fabrication (Scale 1:1)                                                       |           | bms_slave.kicad_pcb |                         | Aidan Brzezinski |             | 2024-04-13    |           | + (Unreleased) |  |
| Sheet Path:                                                                          |           |                     |                         | Reviewer:        |             | Size:         |           | Sheet:         |  |
|                                                                                      |           |                     |                         |                  |             | A4            |           | 2 of 10        |  |

BMS\_SLAVE Fabrication Document

Drill Table

| Symbol   | Count | Hole Size          | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|----------|-------|--------------------|--------|------------|------------------|-----------|
| ×        | 11    | 0,20mm (7,87mils)  | PTH    | Round      | L1 - L4          | Via       |
| ○        | 36    | 0,20mm (8,00mils)  | PTH    | Round      | L1 - L4          | Pad       |
| +        | 1     | 0,30mm (11,81mils) | PTH    | Round      | L1 - L4          | Via       |
| □        | 44    | 1,07mm (42,00mils) | PTH    | Round      | L1 - L4          | Pad       |
| Total 92 |       |                    |        |            |                  |           |

Drill Drawing L1 - L4 (Scale 1:1)



|                                         |                                     |                               |                            |                             |
|-----------------------------------------|-------------------------------------|-------------------------------|----------------------------|-----------------------------|
| Comments:                               | Company:<br>Blue Devil Motor Sports |                               | Variant:<br>PRELIMINARY    | Git Hash:<br>cd3370c        |
|                                         | Board Name:<br>BMS_SLAVE            |                               | Project Name:<br>BMS Slave |                             |
| Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>bms_slave.kicad_pcb   | Designer:<br>Aidan Brzezinski | Date:<br>2024-04-13        | Revision:<br>+ (Unreleased) |
| Sheet Path:                             |                                     | Reviewer:                     | Size:<br>A4                | Sheet:<br>3 of 10           |

# BMS\_SLAVE Fabrication Document

Drill Table

| Symbol   | Count | Hole Size           | Plated | Hole Shape | Drill Layer Pair | Hole Type  |
|----------|-------|---------------------|--------|------------|------------------|------------|
| ×        | 4     | 1.07mm (42,00mils)  | NPTH   | Round      | L1 - L4          | Mechanical |
| ○        | 2     | 2.46mm (97,00mils)  | NPTH   | Round      | L1 - L4          | Mechanical |
| +        | 4     | 3.05mm (120,00mils) | NPTH   | Round      | L1 - L4          | Mechanical |
| □        | 4     | 3.20mm (125,98mils) | NPTH   | Round      | L1 - L4          | Mechanical |
| Total 14 |       |                     |        |            |                  |            |

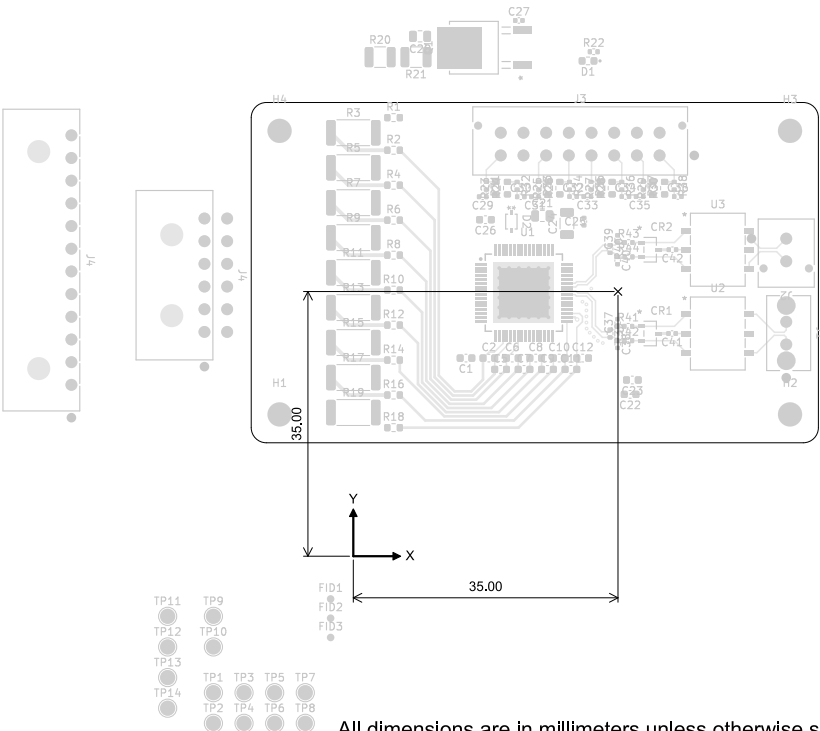
Drill Drawing L1 - L4 (Scale 1:1)



|                                         |                                     |                               |                            |                             |
|-----------------------------------------|-------------------------------------|-------------------------------|----------------------------|-----------------------------|
| Comments:                               | Company:<br>Blue Devil Motor Sports |                               | Variant:<br>PRELIMINARY    | Git Hash:<br>cd3370c        |
|                                         | Board Name:<br>BMS_SLAVE            |                               | Project Name:<br>BMS Slave |                             |
| Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>bms_slave.kicad_pcb   | Designer:<br>Aidan Brzezinski | Date:<br>2024-04-13        | Revision:<br>+ (Unreleased) |
| Sheet Path:                             |                                     | Reviewer:                     | Size:<br>A4                | Sheet:<br>4 of 10           |

BMS\_SLAVE Fabrication Document

Top Test Points (Scale 1:1)



| Ref. | Net      | X [mm] | Y [mm] |
|------|----------|--------|--------|
| TP1  | CVDD     | -5.00  | -33.05 |
| TP2  | DVDD     | -5.00  | -37.10 |
| TP3  | AVDD     | -0.95  | -33.05 |
| TP4  | LDOIN    | -0.95  | -37.10 |
| TP5  | VCOLL    | 3.10   | -33.05 |
| TP6  | LDOIN    | 3.10   | -37.10 |
| TP7  | TSREF    | 7.15   | -33.05 |
| TP8  | GND      | 7.15   | -37.10 |
| TP9  | T_P1     | -5.00  | -22.95 |
| TP10 | GND      | -5.00  | -27.00 |
| TP11 | U1_COMLP | -11.05 | -22.95 |
| TP12 | U1_COMLN | -11.05 | -27.00 |
| TP13 | U1_COMHP | -11.05 | -31.05 |
| TP14 | U1_COMHN | -11.05 | -35.10 |

| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|

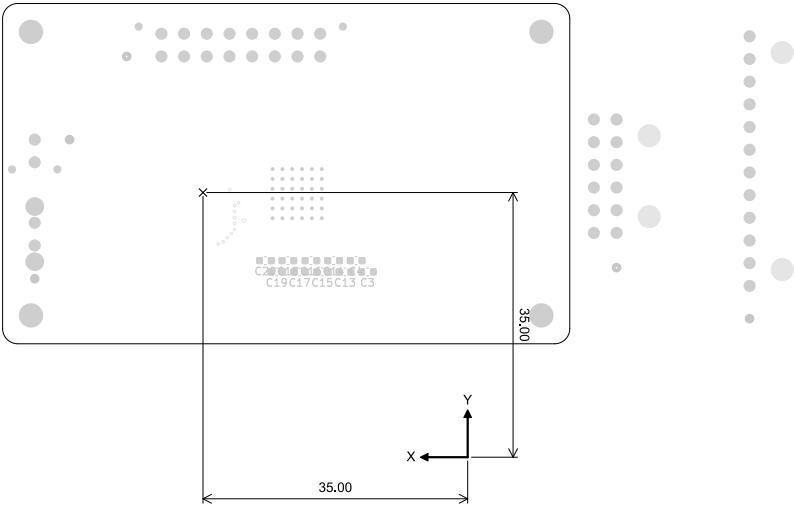
All dimensions are in millimeters unless otherwise specified.

|  |                                             |                                     |                               |                            |                             |
|--|---------------------------------------------|-------------------------------------|-------------------------------|----------------------------|-----------------------------|
|  | Comments:                                   | Company:<br>Blue Devil Motor Sports |                               | Variant:<br>PRELIMINARY    | Git Hash:<br>cd3370c        |
|  | Sheet Title:<br>Top Test Points (Scale 1:1) | Board Name:<br>BMS_SLAVE            |                               | Project Name:<br>BMS Slave |                             |
|  | Sheet Path:                                 | File Name:<br>bms_slave.kicad_pcb   | Designer:<br>Aidan Brzezinski | Date:<br>2024-04-13        | Revision:<br>+ (Unreleased) |
|  |                                             | Reviewer:                           |                               | Size:<br>A4                | Sheet:<br>5 of 10           |

BMS\_SLAVE Fabrication Document

Bottom Test Points (Scale 1:1)

| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|

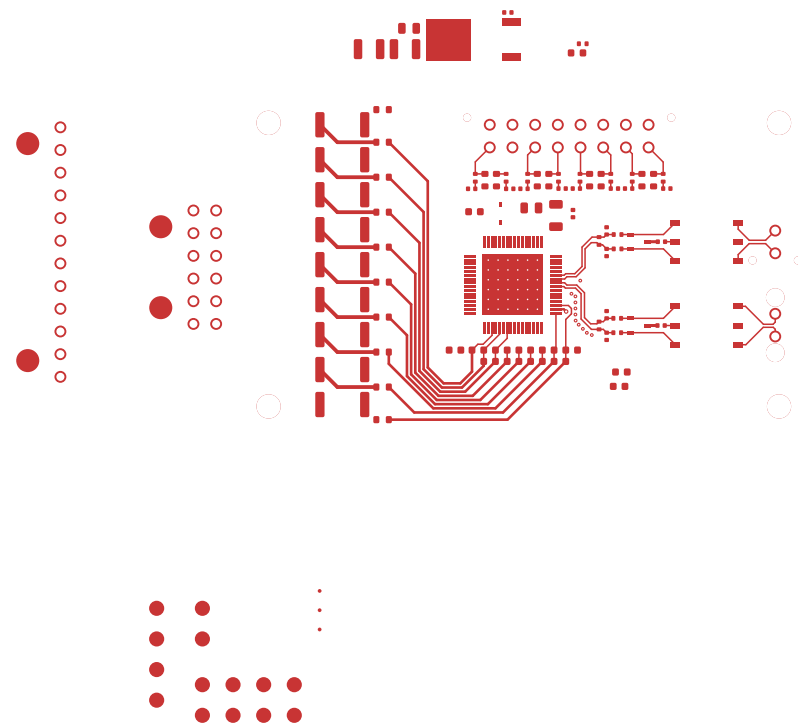


All dimensions are in millimeters unless otherwise specified.

|  |                                                |                                     |                               |                         |                             |
|--|------------------------------------------------|-------------------------------------|-------------------------------|-------------------------|-----------------------------|
|  | Comments:                                      | Company:<br>Blue Devil Motor Sports |                               | Variant:<br>PRELIMINARY | Git Hash:<br>cd3370c        |
|  | Sheet Title:<br>Bottom Test Points (Scale 1:1) | File Name:<br>bms_slave.kicad_pcb   | Designer:<br>Aidan Brzezinski | Date:<br>2024-04-13     | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                    |                                     | Reviewer:                     | Size:<br>A4             | Sheet:<br>6 of 10           |

BMS\_SLAVE Fabrication Document

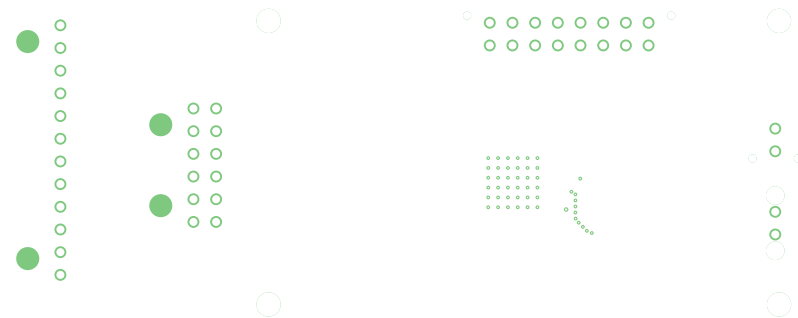
L1 (Scale 1:1)



|                                                                                      |                                |                                     |                               |                         |                             |
|--------------------------------------------------------------------------------------|--------------------------------|-------------------------------------|-------------------------------|-------------------------|-----------------------------|
|  | Comments:                      | Company:<br>Blue Devil Motor Sports |                               | Variant:<br>PRELIMINARY | Git Hash:<br>cd3370c        |
|                                                                                      | Sheet Title:<br>L1 (Scale 1:1) | Board Name:<br>BMS_SLAVE            | Project Name:<br>BMS Slave    |                         |                             |
|                                                                                      | Sheet Path:                    | File Name:<br>bms_slave.kicad_pcb   | Designer:<br>Aidan Brzezinski | Date:<br>2024-04-13     | Revision:<br>+ (Unreleased) |
|                                                                                      |                                |                                     | Reviewer:                     | Size:<br>A4             | Sheet:<br>7 of 10           |

BMS\_SLAVE Fabrication Document

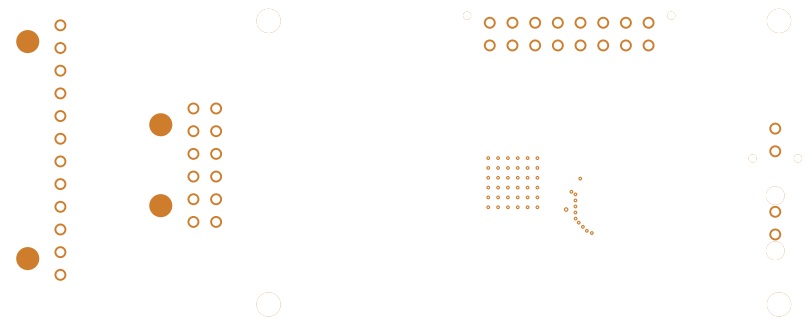
L2 (Scale 1:1)



|                                                                                      |                                |                                     |                               |                         |                             |
|--------------------------------------------------------------------------------------|--------------------------------|-------------------------------------|-------------------------------|-------------------------|-----------------------------|
|  | Comments:                      | Company:<br>Blue Devil Motor Sports |                               | Variant:<br>PRELIMINARY | Git Hash:<br>cd3370c        |
|                                                                                      | Sheet Title:<br>L2 (Scale 1:1) | File Name:<br>bms_slave.kicad_pcb   | Designer:<br>Aidan Brzezinski | Date:<br>2024-04-13     | Revision:<br>+ (Unreleased) |
|                                                                                      | Sheet Path:                    |                                     | Reviewer:                     | Size:<br>A4             | Sheet:<br>8 of 10           |

BMS\_SLAVE Fabrication Document

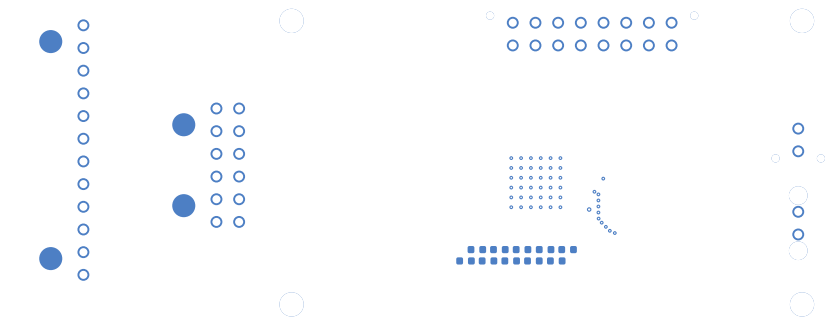
L3 (Scale 1:1)



|                                                                                      |                                |                                     |                               |                            |                             |
|--------------------------------------------------------------------------------------|--------------------------------|-------------------------------------|-------------------------------|----------------------------|-----------------------------|
|  | Comments:                      | Company:<br>Blue Devil Motor Sports |                               | Variant:<br>PRELIMINARY    | Git Hash:<br>cd3370c        |
|                                                                                      | Sheet Title:<br>L3 (Scale 1:1) | Board Name:<br>BMS_SLAVE            |                               | Project Name:<br>BMS Slave |                             |
|                                                                                      | Sheet Path:                    | File Name:<br>bms_slave.kicad_pcb   | Designer:<br>Aidan Brzezinski | Date:<br>2024-04-13        | Revision:<br>+ (Unreleased) |
|                                                                                      |                                | Reviewer:                           |                               | Size:<br>A4                | Sheet:<br>9 of 10           |

BMS\_SLAVE Fabrication Document

L4 (Scale 1:1)



|                                                                                      |                                |                                     |                               |                         |                             |
|--------------------------------------------------------------------------------------|--------------------------------|-------------------------------------|-------------------------------|-------------------------|-----------------------------|
|  | Comments:                      | Company:<br>Blue Devil Motor Sports |                               | Variant:<br>PRELIMINARY | Git Hash:<br>cd3370c        |
|                                                                                      | Sheet Title:<br>L4 (Scale 1:1) | File Name:<br>bms_slave.kicad_pcb   | Designer:<br>Aidan Brzezinski | Date:<br>2024-04-13     | Revision:<br>+ (Unreleased) |
|                                                                                      | Sheet Path:                    |                                     | Reviewer:                     | Size:<br>A4             | Sheet:<br>10 of 10          |